نام و تخلص : داود اندیشمند
ولد : صادق
دیپارتمنت : انجینیری نرم افزار
حل سیزده سوال جاوا چیپتر یک
سوال یک:

```
public class main {
    public static void main(String[]args) {
        System.out.println("welcome to java");
        System.out.println("welcome to computer science");
        System.out.println("programming is fun");
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
welcome to java
welcome to computer science
programming is fun
```

سوال دو:

```java
public class main {
    public static void main(String[]args) {
        System.out.println("welcome to java");
        System.out.println("welcome to java");
        System.out.println("welcome to java");
        System.out.println("welcome to java");
        System.out.println("welcome to java");
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
welcome to java
welcome to java
welcome to java
welcome to java
welcome to java
```

سوال سه:

```
public class main {
    public static void main(String[]args) {
        System.out.println("     J     A      V     V     A");
        System.out.println("     J    A A      V   V     A A");
        System.out.println(" J   J   AAAAA      V V     AAAAA");
        System.out.println("  J J   A     A      V     A     A");
    }
}
```

```
     J     A      V     V     A
     J    A A      V   V     A A
 J   J   AAAAA      V V     AAAAA
  J J   A     A      V     A     A
```

سوال چهار:

```
public class main {
    public static void main(String[]args) {
        System.out.println("a     a^2     a^3");
        System.out.println("1     1       1");
        System.out.println("2     4       8");
        System.out.println("3     9       27");
        System.out.println("4     16      64");
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
a     a^2     a^3
1     1       1
2     4       8
3     9       27
4     16      64
```

سوال پنج:

```
public class main {
    public static void main(String[]args) {
        System.out.println((9.5*4.5-2.5*3)/(45.5-3.5));

    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
0.8392857142857143
```

سوال شش:

```
public class main {
    public static void main(String[]args) {
        System.out.println(1+2+3+4+5+6+7+8+9);

    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
45
```

سوال هفتم: پیدا کردن قیمت π

```
public class main {
    public static void main(String[]args) {
        System.out.println(4*(1.0 -1.0/3 +1.0/5 -1.0/7 +1.0/9 -1.0/11));
        System.out.println(4*(1.0 -1.0/3 +1.0/5 -1.0/7 +1.0/9 -1.0/11 +1.0/13));
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
2.9760461760461765
3.2837384837384844
```

سوال هشتم:

```
public class main {
    public static void main(String[]args) {
        double radious = 5.5;
        double pi = 3.14159;
        double perimeter = 2 * radious * pi;
        double area = radious * radious * pi;
    System.out.println("perimeter="+ perimeter);
    System.out.println("area="+ area);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
perimeter=34.55749
area=95.0330975
```

سوال نهم:

```
public class main {
    public static void main(String[]args) {
        double height = 7.9;
        double width = 4.5;
        double perimeter = (2 *( width + height));
        double area = width * height;
    System.out.println("perimeter="+ perimeter);
    System.out.println("area="+ area);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
perimeter=24.8
area=35.550000000000004
```

سوال دهم:

```
public class main {
    public static void main(String[]args) {
        double kilometers = 14;
        double miles = kilometers/1.6;
        double hours = ((45.0 + 30.0/60)/60);
        double speed = miles/hours;
    System.out.println("average speed in miles per hour=" + speed);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Documents>java main
average speed in miles per hour=11.538461538461538
```

سوال یازدهم:

```
public class main {
    public static void main(String[]args) {
        int population = 312032486;
        int secondsinyear=365*24*60*60;
        int births = secondsinyear /7;
        int deaths = secondsinyear /13;
        int immigrants = secondsinyear /45;
        population = population + births - deaths + immigrants;
        System.out.println("population after 1 year=" + population);
        population = population + births + immigrants - deaths;
        System.out.println("population after 2 year =" + population);
        population = population + births + immigrants -deaths;
        System.out.println("population after 3 year = =" +population);
        population = population + births + immigrants - deaths;
        System.out.println("population after 4 year =" + population);
        population = population + births + immigrants - deaths;
        System.out.println("population after 5 year =" + population);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Desktop>java main
population after 1 year=314812582
population after 2 year =317592678
population after 3 year = =320372774
population after 4 year =323152870
population after 5 year =325932966
```

سوال دوازدهم :

```
public class main {
    public static void main(String[]args) {
        double mile = 24;
        double kilometer = mile * 1.6;
        double hour = (1 + 40/60 + 35/3600);
        double speed = kilometer/hour;
        System.out.print("average speed =" + speed);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Desktop>java main
average speed =38.400000000000006
```

سوال سیزده:

```
public class main {
    public static void main(String[]args) {
        double a = 3.4;
        double b = 50.2;
        double e = 44.5;
        double c = 2.1;
        double d = 0.55;
        double f =5.9;
        double x = (a*d - b*f)/(a*d - b*c);
        double y = (a*f - e*c)/(a*d - b*c);
        System.out.println("x =" + x);
        System.out.print("y =" + y);
    }
}
```

```
C:\Users\PREMIER COMPUTER\Desktop>java main
x =2.842201834862385
y =0.7087397392563978
```